Answer Key - Chemistry - Undergraduate Level Assessment 9 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 101.1 x 10^7 T-1 s-1; B) 26.75 x 10^7 T-1 s-1; C) 7.081 x 10^7 T-1 s-1; D) 0.9377 x 10^7 T-1 s-1
   Note: The magnetogyric ratio is calculated by dividing the NMR frequency (198.52 MHz) by the magnetic field strength (750 MHz), yielding a value of approximately 7.081 x 10^7 T^-1s^-1.

1. **Answer:** D
   Options: A) T_1, unlike T_2, is sensitive to very low-frequency molecular motions.; B) T_2, unlike T_1, is sensitive to very low-frequency molecular motions.; C) T_1, unlike T_2, is sensitive to molecular motions at the Larmor frequency.; D) T_2, unlike T_1, is sensitive to molecular motions at the Larmor frequency.
   Note: The correct answer is based on the fact that T_1 (spin-lattice relaxation time) is primarily influenced by energy transfer to the surrounding lattice, while T_2 (spin-spin relaxation time) is affected by interactions between nuclear spins and molecular motions at the Larmor frequency, which can lead to faster relaxation and potentially shorter T_2 values compared to T_1.

2. **Answer:** D
   Options: A) I only; B) II only; C) III only; D) I and III only
   Note: The normal modes of a carbon dioxide molecule that are infrared-active must involve a change in the dipole moment, which occurs in the asymmetric stretching and bending modes, but not in the symmetric stretching mode, as it does not produce a net change in the dipole moment.

3. **Answer:** A
   Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays
   Note: Cobalt-60 is produced through a nuclear reaction where cobalt-59 absorbs a neutron, leading to its transformation into cobalt-60, which is utilized in radiation therapy for cancer treatment.

4. **Answer:** C
   Options: A) Nuclear magnetic resonance; B) Infrared; C) Raman; D) Ultraviolet-visible
   Note: Raman spectroscopy is a light-scattering technique because it involves the inelastic scattering of monochromatic light, typically from a laser, which results in a shift in energy corresponding to vibrational modes of molecules.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because the hydrogen atom in CHCl 3 undergoes rapid intermolecular exchange due to hydrogen bonding, leading to averaged chemical environments that typically result in a single peak in its spectrum rather than distinct peaks.

7. **Answer:** True
   Options: A) True; B) False
   Note: At standard temperature and pressure, bromine (Br_2) is a diatomic molecule that exhibits a red-brown color and is indeed a volatile liquid due to its relatively low boiling point of 58.8°C.

8. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the 55 Mn nuclide, being a stable isotope with specific nuclear configurations, has three distinct energy levels corresponding to its nuclear spin states that can be populated under certain conditions.

9. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because gadolinium (Gd) has the electron configuration [Xe] 4 f^7 5 d^1, indicating that it does possess a partially filled 4 f subshell with seven electrons.

10. **Answer:** False
    Options: A) True; B) False
    Note: The equilibrium polarization of 13 C nuclei in a 20.0 T magnetic field at 300 K is actually lower than 10.8 x $10^-5 due to the much smaller magnetic moment of 13C compared to protons, res$

Long Form Response

11. **Answer:** The stability of the +1 and +3 oxidation states in group 13 elements varies, with thallium (Tl) exhibiting a notable preference for the +1 state. This preference is largely attributed to relativistic effects, which play a significant role in heavier elements like thallium. In the +1 oxidation state, thallium achieves a stable electron configuration similar to that of the noble gas xenon, while in the +3 state, it experiences increased ionization energy and lower stability due to the higher oxidation state involving the loss of more electrons. Additionally, the +1 state is more stabilized by the inert pair effect, where the s-electrons are held more tightly and are less likely to participate in bonding, thus reinforcing the stability of the +1 oxidation state over the +3 state in thallium.
    Note: correct because it identifies that thallium's preference for the +1 oxidation state

is influenced by relativistic effects, resulting in a stable electron configuration akin to xenon, while the +3 state is less stable due to higher ionization energy and increased oxidation challenges.

12. **Answer:** In the reaction 3 Cl-(aq) + 4 CrO 4^2-(aq) + 23 H+(aq) $\rightarrow$ 3 HClO 2(aq) + 4 Cr_3+(aq) + 10 H_2O(l), Cl-(aq) acts as a reducing agent by donating electrons to the chromium species. As Cl- is oxidized to HClO 2, it loses electrons, which facilitates the reduction of CrO 4^2- to Cr_3+. This electron transfer is a defining characteristic of reducing agents, as they enable other substances to gain electrons while undergoing oxidation themselves. Therefore, the role of Cl-(aq) as a reducing agent is crucial in driving the redox chemistry of the reaction forward.
    Note: correct because it accurately describes how Cl-(aq) donates electrons during the reaction, thus undergoing oxidation to HClO 2 while facilitating the reduction of CrO 4^2- to Cr_3+, which exemplifies the defining characteristic of a reducing agent.

End of Answer Key